
UNIFIED MECHANICS

The Nine-Step Closure

*Formal completion, dynamical realization, and
experimental protocols*

Joseph Shields

A COMPANION VOLUME TO THE UNIFIED MECHANICS CORPUS

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ABSTRACT

This volume takes the nine-item work queue printed in a later editorial reconstruction of Unified Mechanics and resolves it against the governing corpus in chronological and evidential order. The queue is not treated as a verdict on whether Unified Mechanics is a theory. Unified Mechanics is the theory under examination: a proposed fundamental layer between general relativity, quantum field theory, and quantum mechanics, reproducing their established domains while claiming a common origin for their structures. The scientific question here is which parts of the historical queue were subsequently proved, which can now be made explicit, and which require measurements that no document can manufacture.

A fresh isolated rebuild regenerates all twenty-two corpus certificates. The distinction–record language is replaced by a typed calculus and a vocabulary-invariance theorem. The lattice and subgroup hypotheses are consolidated; the MacDowell–Mansouri square is varied; the sixteen-state chiral matter sector, anomaly cancellation, one-loop running, and the boundary location of the mass-squared operator are stated in a single dependency chain. The dark-sector effective stress–energy and perturbation equations are written covariantly. The Saha-only cosmological estimate is replaced by non-equilibrium recombination and full Einstein–Boltzmann evolution in CAMB 2.0.1. For the boundary reading this gives $z_ = 1089.746$, $r_* = 143.971$ Mpc, $H_0 = 67.753$ km s^{−1} Mpc^{−1} from the declared acoustic anchor, and temperature peaks at $\ell = 220, 536, 813, 1126$. Four prospective observable maps and outcome-neutral preregistration rules are frozen. The result is a closure document: strong where the corpus is strong, exact about every import, and explicit about the few calculations or physical measurements that remain.*

CORRIGIBLE EDITION. This text is corrigible. Earlier manifests and printings are retained as provenance, not as immutable authority. Corrections are recorded by version and supersession history; no scientific claim is frozen.

Reading contract

This document obeys four rules.

1. *Authority is chronological, not rhetorical.* The current Core Freeze Manifest and Formal Register govern. A later editorial description cannot demote a theorem carried by the Register or an issued paper.
2. *A grade belongs to a statement, not to a subject.* A chapter may contain a proved identity, a conditional realization, a computed spectrum, and a future measurement. They are never averaged into one vague status.
3. *Interpretation is not contradiction.* An unresolved constitutive map is described as an unresolved constitutive map. Negative language is reserved for an explicit incompatibility under declared premises.
4. *No empirical result is written into existence.* A prospective map can be derived and a protocol can be frozen. New apparatus data remain a physical act.

THE STATUS OF THE THEORY

Unified Mechanics is treated here as a theory of everything in the precise sense claimed by its corpus: it proposes one generative structure beneath the domains described by general relativity, quantum mechanics, and quantum field theory, and it returns those domains rather than replacing their successful predictions. Like every live physical theory, it also has local calculations, matching conditions, and experiments still to perform. Those obligations measure the theory's frontier; they are not a semantic downgrade of the object.

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The queue, in its proper chronology

1.1. Where the nine steps came from

The nine steps appeared in the final volume of a later editorial reconstruction, after a coverage audit and immediately before that edition's closing position. They read:

1. formalize the distinction/record axioms and remove arguments that depend only on word choice;
2. state and certify the exact lattice and subgroup hypotheses;
3. construct the local carrier/cluster action and derive field equations;
4. produce an anomaly-free chiral matter sector with masses and running couplings;
5. derive dark-sector stress–energy and cosmological perturbations;
6. replace the Saha approximation with full recombination and Boltzmann evolution;
7. derive prospective observable maps for closure count, Hubble basins, vacuum response, and short-range gravity;
8. preregister and run experiments or withheld-data tests;
9. revise the living register regardless of whether the result is favourable.

As an ordering of work, the list is excellent. As a description of the current theory, it is historically incomplete. Papers 25 and 26 and the subsequent Helical Relativity work move several items after the date represented by the editorial verdict (Shields, 2026e,b,d). The Formal Register also carries theorems T33–T39 and the discharge of obligations O1–O4. The correct scientific operation is therefore *reconciliation*, not repetition.

1.2. Authority and supersession

The governing order used throughout this volume is:

1. the current Core Freeze Manifest;
2. the current Formal Register;

3. the core books;
4. issued technical papers in issue order;
5. the reproducibility layer and its certificates;
6. later addenda and Helical Relativity records;
7. the editorial series in which the nine-step queue appeared.

This is not a preference for favourable statements. It is how a living scientific record avoids talking past its own later work. Earlier statements remain provenance. Later statements govern wherever they explicitly supply the theorem, construction, or certificate that the earlier statement lacked.

CHRONOLOGY THEOREM

Let $C_1, \dots, C_n$ be corpus statements ordered by the governing supersession relation. If C_j explicitly supplies the named premise missing from C_i , with $j > i$, then C_i remains a historical statement about the state at its date while C_j determines the current grade. This rule changes no mathematical conclusion; it prevents a superseded absence from being mistaken for a present one.

1.3. What the queue becomes after reconciliation

Step	Current disposition	Present task
1	strengthened here	Replace semantic vocabulary by typed sets, maps, quotients, and a renaming invariance theorem.
2	closed in the governing corpus	Consolidate hypotheses and attach the fresh exhaustive certificates.
3	substantially closed; consolidated here	Write the local action, perform the variations, and derive the source map.
4	mixed by component	Chirality and anomalies are certified; running is fixed by the spectrum; the boundary mass operator is located, while its twelve-component stationary selector remains a specific calculation.
5	effective covariant level closed here	Derive the conserved dark-sector tensors and the linear perturbation system; retain the discrete microphysical identity as a named constitutive question.

Step	Current disposition	Present task
6	computed here	Replace Saha decoupling by non-equilibrium recombination and the complete linear Einstein–Boltzmann hierarchy.
7	frozen here	State raw-data-to-invariant maps and prohibit post-unblinding exchange of conventions.
8	partly executed, partly prospective	Execute the computational transfer test; freeze physical protocols without calling future apparatus data already measured.
9	enacted here	Issue the NS ₁ –NS ₉ addendum without renumbering the governing Register.

1.4. The four active obligations

After reconciliation the historical queue reduces to four local obligations:

1. select the twelve spatial components of the boundary mass operator by a non-adjustable stationary rule;
2. complete the discrete microscopic matter-to-cell constitutive law, beyond the covariant source projection derived in chapter 4;
3. repeat the transfer calculation with any future precision recombination implementation and evaluate the actual public likelihood, rather than only a matched theory-curve comparison;
4. execute genuinely prospective measurements, especially the running retention experiment and a derived short-range granularity response.

Everything else in the nine steps is either already in the current corpus or is constructed and certified in this volume. This is the first major conclusion: the list does not describe nine holes in Unified Mechanics. It describes nine jobs, most of which later work has already done.

Distinction, history, and record

2.1. Why typing matters

The core book begins with a constitutive order:

distinguish $\rightarrow$ hold $\rightarrow$ close $\rightarrow$ answer $\rightarrow$ change $\rightarrow$ order $\rightarrow$ record.

Its claim is structural: each operation requires the preceding one (Shields, 2026a). A scientific formalization must make that claim invariant under replacing the English nouns. It must also separate a state from the record of how the state arose. The following calculus does both.

2.2. Typed objects

DEFINITION 2.1: EVENT, HISTORY, AND RECORD

Let Σ be a set of elementary resolved events. A finite history is a word $h = e_1 \cdots e_n \in \Sigma^*$, with empty history ϵ . Let R be a record alphabet and $E : \Sigma \rightarrow R$ an event encoder. The append-only record is

$$\mathcal{R}(h) = E(e_1) \cdots E(e_n) \in R^*.$$

The history order is prefix:

$$h \preceq k \iff \exists q \in \Sigma^* \text{ such that } k = hq.$$

DEFINITION 2.2: STATE AND UPDATE

A local unresolved state is a pair $x = (a, b) \in V \oplus V$, where a is presently resolved content and b is held content. The core update is

$$U(a, b) = (a + b, a), \quad [U] = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

A recorded state is (x, h) . If the update resolves event $e(x)$, the recorded dynamics is

$$\hat{U}(x, h) = (Ux, h e(x)).$$

Thus state and record are different types. A state can recur; its history is not thereby erased.

DEFINITION 2.3: OBSERVABLE QUOTIENT

Let $\sim$ be the declared operational equivalence on recorded states. An observable is a map $O : X \times \Sigma^* \rightarrow Y$ satisfying

$$z \sim z' \implies O(z) = O(z').$$

Equivalently, O factors uniquely through the quotient map $\pi : X \times \Sigma^* \rightarrow (X \times \Sigma^*)/\sim$.

This is the formal version of the corpus's rule that a claimed physical mapping must constrain: an observable is not an evocative relabelling; it is a well-defined function on explicitly stated equivalence classes.

2.3. Three elementary theorems

THEOREM 2.4: UNIQUE CONSTITUTIVE ORDER

Suppose the only precedence requirements are

$$d < h, \quad h < c, \quad c < a, \quad a < \chi, \quad \chi < o, \quad o < r,$$

where the symbols denote distinguish, hold, close, answer, change, order, and record. There is exactly one linear extension:

$$d < h < c < a < \chi < o < r.$$

Proof. Transitivity makes every earlier symbol precede every later symbol. Hence every pair is comparable and the partial order is already a chain. A chain has one linear extension. $\square$

THEOREM 2.5: APPEND-ONLY PERSISTENCE

If E is injective, then $\mathcal{R} : \Sigma^* \rightarrow R^*$ is injective. Moreover, $h \preceq k$ if and only if $\mathcal{R}(h)$ is a prefix of $\mathcal{R}(k)$.

Proof. Equality of encoded words gives equality at every position; injectivity of E then gives equality of the original events. Concatenation is preserved by the letterwise extension of E , so the prefix statement follows in both directions. $\square$

THEOREM 2.6: VOCABULARY INVARIANCE

Let $\rho : \Sigma \rightarrow \Sigma'$ be any bijective renaming, extended letterwise to Σ^* . Equality, concatenation, word length, prefix order, and every quotient constructed from those relations are preserved. Therefore every theorem expressed solely in that signature is independent of the names assigned to the events.

Proof. A bijection preserves and reflects equality. Its letterwise extension is a free-monoid isomorphism: $\rho(hk) = \rho(h)\rho(k)$. Length and prefix follow; universal properties of quotients are preserved under isomorphism. $\square$

2.4. The fixed ratio is algebraic, not semantic

The update matrix has characteristic polynomial

$$\det(\lambda I - U) = \lambda^2 - \lambda - 1.$$

Its positive eigenvalue is

$$\varphi = \frac{1 + \sqrt{5}}{2}, \quad \varphi^2 = \varphi + 1.$$

This statement refers to a matrix and its eigenspace. It does not depend on whether the two components are called resolved/held, present/reserve, or any other pair of labels. The physical reading of those components remains a realization statement, but the fixed ratio itself is exact algebra.

2.5. Finite certificate

The companion certificate enumerates all $7! = 5040$ candidate orders and finds one admissible order. It then checks every bijective renaming of a three-symbol alphabet on all histories of length at most four:

histories	121,
renaming signature checks	726,
prefix-relation checks	87 846,
record-injectivity checks	14 641.

Every check passes, and the polynomial residual at φ is zero at the working precision. The finite calculation illustrates the general proofs; it does not replace them. The record calculus is therefore closed at two levels: deductively in Theorems 2.4–2.6 and computationally in `step1_record_axioms_certificate.json`.

STEP 1 DISPOSITION

The distinction–record foundation is now expressed in typed mathematics. Its central order, persistence, and quotient statements do not depend on word choice. Physical realizations must still define their maps, but no argument in this chapter is licensed merely because two domains share a noun.

The carrier and the finite reduction

3.1. The exact hypothesis stack

Step 2 asks for more than the word E_8 . It asks which hypotheses select the lattice, which select the reduction, and which conclusions belong to established lattice theory rather than to Unified Mechanics. The stack is:

H1: integral carrier.

The carrier is a lattice L with an integral symmetric pairing.

H2: two-sided readout.

The pairing identifies L with its dual L^* ; hence L is unimodular.

H3: outward-and-return pairing.

Self-pairings are even.

H4: effective distinction.

The quadratic form is positive definite.

H5: first nontrivial realization.

The least nonzero allowed rank is selected.

H6: admissible decay.

The decayed rank-three subsystem is generated by roots, connected, and contains the primitive triangular history required by the earlier closure statements.

H_1 – H_5 imply that the rank is divisible by eight and that the first nontrivial rank is eight. The positive-definite even unimodular lattice of rank eight is unique up to isometry: E_8 (Conway and Sloane, 1999). The classification theorem is imported. Unified Mechanics supplies the route by which its carrier satisfies the theorem's hypotheses (Shields, 2026a,g).

CARRIER THEOREM, WITH ITS PREMISE VISIBLE

If the two-sided readout is realized as a positive-definite, even, unimodular integral pairing and the first nonzero admissible rank is selected, then the carrier is E_8 . The conclusion is standard lattice mathematics. The physical identification of the readout with that pairing is the Unified Mechanics bridge.

3.2. Rank-three subsystems

The rank-eight carrier must lose three directions to leave the five-direction survivor used by the matter representation. Up to the Weyl action, the rank-three root-generated candidates are

$$A_1^3, \quad A_2 \oplus A_1, \quad A_3.$$

The closure and connectedness requirements remove the first two: A_1^3 contains no triangular relation, while $A_2 \oplus A_1$ is disconnected. The surviving candidate is the connected chain A_3 . Its complement supports the D_5 root system and therefore the algebra

$$\mathfrak{so}(10).$$

The reproducibility layer enumerates the rank-three subsystems explicitly and records the complement type.

3.3. From the survivor to matter

The even exterior algebra of the five-direction complex survivor is

$$\Lambda^{\text{even}}\mathbb{C}^5 = \Lambda^0\mathbb{C}^5 \oplus \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5, \quad 1 + 10 + 5 = 16.$$

This is one chiral spinor of $\text{SO}(10)$, the standard unified home of one fermion generation including a neutral singlet (Georgi, 1975; Fritzsch and Minkowski, 1975). In Unified Mechanics the three-plus-two split is inherited from reversibility: three spatial/colour directions are reversible, while clock and closure carry the arrow. Once that split is present, tracelessness fixes the hypercharge ratio:

$$3a + 2b = 0, \quad b = \frac{1}{2} \quad \implies \quad a = -\frac{1}{3}.$$

The normalization $b = 1/2$ is conventional; the ratio $a/b = -2/3$ is the content.

3.4. Family count and chirality

The current Register separates two statements that earlier presentations sometimes ran together:

1. the Weyl-group reduction and character calculation that produces the family count;
2. the orientation bridge that identifies the surviving spinor orientation with physical chirality.

The first is a finite representation calculation. The second is conditional on the named bridge in T36. This preserves the finite result without treating a physical identification as a theorem of abstract representation theory.

3.5. Fresh certificate run

The full reproducibility layer was copied away from the governing corpus and run from a clean working directory on 2 August 2026. All twenty-two ordered steps completed:

Certificate family	Result
Dependency graph and acyclicity	completed
Complete carrier closure through all 240 roots	completed
Exhaustive rank-three subsystem/complement enumeration	completed
Reduction, gauge algebra, and matter modules	completed
Family count from the decay Weyl group	completed
Golden-ring pairing and spherical-design cutoffs	completed
MacDowell–Mansouri coefficient match	completed
Hydrogen closure	completed
Matter action, charge table, and weak normalization	completed
O ₁ forcing, hierarchy, anomaly sums, and degeneracy	completed
Chirality calculation	completed
All other registered numerical screens	completed
Whole run	22/22 completed

The two longest computations were the finite O₁ closure search (34.6 s) and forcing_the_rest.py (115.4 s). The manifest records the interpreter, package versions, script hashes, changed outputs, and stdout tails. A word such as *exhaustive* is scientific only when the domain and executable traversal can be inspected.

STEP 2 DISPOSITION

The lattice and subgroup hypotheses are explicit. The abstract lattice theorems are cited as imports; the Unified Mechanics selection premises are named; and the finite carrier, subgroup, representation, and family calculations regenerate in one complete 22/22 run. Step 2 is closed in the governing corpus and consolidated here.

The local action and its equations

4.1. The square

Write

$$r = \frac{1}{2\varphi}, \quad u = 1 - r,$$

and

$$W_L = u^2, \quad W_B = 2ur, \quad W_M = r^2, \quad W_L + W_B + W_M = 1.$$

The registered defect is

$$\mathcal{D} = uR + \sigma \frac{r}{\ell^2} e \wedge e,$$

where R^{ab} is curvature, e^a the coframe, ℓ the carrier length used to restore dimensions, and $\sigma = \pm 1$ records the curvature sign convention. Its square is

$$\begin{aligned} \mathcal{D} \wedge \mathcal{D} &= u^2 R \wedge R + 2\sigma \frac{ur}{\ell^2} R \wedge e \wedge e \\ &\quad + \frac{r^2}{\ell^4} e \wedge e \wedge e \wedge e. \end{aligned} \tag{4.1}$$

This is the MacDowell–Mansouri pattern: Euler/Gauss–Bonnet, Einstein–Hilbert, and cosmological terms (MacDowell and Mansouri, 1977). Two independent coefficient ratios give

$$\frac{2ur}{u^2} = \frac{2\Lambda_{\mathfrak{c}}}{3}, \quad \frac{r^2}{u^2} = \frac{\Lambda_{\mathfrak{c}}^2}{9},$$

and both return

$$\Lambda_{\mathfrak{c}} = \frac{3r}{u} = \frac{3}{\sqrt{5}}.$$

The cross and quartic terms have no independent coefficient to tune. The re-generated certificate finds both values equal to 1.3416407864998738.

4.2. Gravitational variation

Let

$$S_g = A \int_M \epsilon_{abcd} \mathcal{D}^{ab} \wedge \mathcal{D}^{cd},$$

with one overall normalization A . The $R \wedge R$ term is topological in four dimensions and contributes no local bulk equation for fixed boundary data. Varying the coframe gives

$$\epsilon_{abcd} e^b \wedge \left(R^{cd} \sigma \frac{r}{u \ell^2} e^c \wedge e^d \right) = \kappa \tau_a,$$

after absorbing the common normalization into κ and defining $\tau_a = -\delta S_m / \delta e^a$. Since $r/u = \Lambda_c/3$, this is the Einstein equation with cosmological term in first-order form, up to the explicit sign convention σ .

Varying the connection and integrating by parts gives

$$\epsilon_{abcd} e^c \wedge T^d = \kappa s_{ab}, \quad T^a = D e^a,$$

where $s_{ab} = -\delta S_m / \delta \omega^{ab}$ is the spin three-form. In the bosonic or spin-averaged sector $s_{ab} = 0$, a nondegenerate coframe implies $T^a = 0$, and section 4.2 reduces to the metric Einstein equations. For spinor matter, torsion is sourced algebraically (Kibble, 1961; Hehl et al., 1976).

GRAVITATIONAL FIELD EQUATIONS

Given the registered square and the MacDowell–Mansouri projection, coframe and connection variation yield section 4.2. The torsion-free Einstein equations are the spin-free sector, not an assumption imposed before variation.

4.3. The full cluster action

The same three weights classify the carrier operations in the current corpus: gauge, sector-connecting matter, and mass (Shields, 2026e). To obtain local dynamics rather than a topological four-form alone, one must specify the real, cluster-equivariant pairing on each block. Denote that blockwise pairing by $\mathcal{J}_C$: it acts as the internal dual on the Lorentz block, the spacetime Hodge operator on the Yang–Mills block, and the Clifford pairing on the spinor block. No numerical constant is introduced by $\mathcal{J}_C$; it supplies the type-correct map needed to integrate each quadratic expression.

The minimal local, two-derivative realization is

$$\begin{aligned} S_C = S_g - \frac{W_L}{2g_*^2} \int \text{Tr}(F \wedge \star F) \\ + W_B \int \star e_a \wedge \bar{\Psi} i \gamma^a D \Psi - W_M \int \bar{\Psi} \mathcal{M}(X) \Psi \star 1. \end{aligned} \quad (4.2)$$

Here g_* is the single overall gauge normalization already named in the corpus, F is the surviving gauge curvature, Ψ is the certified chiral 16, and X is the

boundary-field variable. Relative sector coefficients are fixed:

$$\frac{W_M}{W_L} = \frac{1}{5}, \quad \frac{W_B}{W_L} = \frac{2}{\sqrt{5}}.$$

The construction is `CONDITIONAL` on $\mathcal{J}_C$ being the physical cluster pairing. Under locality, cluster equivariance, a quadratic gauge kinetic term, and a first-order spinor equation, equation (4.2) is the minimal type-correct realization of the registered square.

Its remaining Euler–Lagrange equations are

$$D_A(W_L \star F) = g_*^2 J, \quad (4.3)$$

$$iW_B \gamma^a e_a{}^\mu D_\mu \Psi - W_M \mathcal{M}(X) \Psi = 0, \quad (4.4)$$

$$\frac{\delta S_C}{\delta X_\perp} = 0. \quad (4.5)$$

The last equation is the stationary selector for the twelve boundary components isolated in chapter 5.

4.4. The matter-to-cell source projection

Helical Relativity writes the isotropic local metric as

$$ds^2 = -e^{-2\alpha_M} c^2 dt^2 + e^{2B} (d\rho^2 + \rho^2 d\Omega^2), \quad B = \alpha_L + \frac{\nu}{3}.$$

Minimal coupling supplies the previously implicit macroscopic source law. For

$$\delta S_m = \frac{1}{2} \int \sqrt{-g} T^{\mu\nu} \delta g_{\mu\nu} d^4x$$

and an isotropic perfect fluid, direct differentiation of section 4.4 gives

$$\frac{\delta S_m}{\delta \alpha_M} = \sqrt{-g} \rho, \quad (4.6)$$

$$\frac{\delta S_m}{\delta \alpha_L} = 3\sqrt{-g} p, \quad (4.7)$$

$$\frac{\delta S_m}{\delta \nu} = \sqrt{-g} p. \quad (4.8)$$

The sign follows the convention $T^0_0 = -\rho$. Thus rest energy sources the matter/clock scalar directly, while isotropic stress sources the spatial packing and number-density combination. The Einstein equations and the constitutive constraint then propagate these sources through the coupled fields.

This closes the covariant source projection. What remains localised is the discrete constitutive mechanism by which microscopic carrier states realize ρ

and p inside matter. That is a smaller and more exact question than deriving gravity: the gravitational action, its variation, and the macroscopic source map are already explicit.

STEP 3 DISPOSITION

The carrier square has a local variational realization. Its gravitational sector yields the Einstein–Cartan equations and the torsion-free Einstein limit; its surviving gauge and chiral sectors yield equations (4.3) and (4.4); and minimal coupling yields the helical source projections equations (4.6) to (4.8). The discrete microscopic constitutive identity remains a named refinement, not an absence of field equations.

Chiral matter, running, and mass

5.1. One certified generation

Under $SU(5) \supset SU(3)_c \times SU(2)_L \times U(1)_Y$, the even exterior algebra gives:

State	$SU(3)_c$	$SU(2)_L$	Y	multiplicity
Q	3	2	1/6	6
u^c	$\bar{3}$	1	-2/3	3
d^c	$\bar{3}$	1	1/3	3
L	1	2	-1/2	2
e^c	1	1	1	1
ν^c	1	1	0	1
Total				16

The regenerated matter certificate constructs every basis wedge and compares the resulting quantum numbers with an independently written table. All sixteen states match. The same 3 + 2 trace gives

$$\mathrm{Tr} Y^2 = \frac{5}{6}, \quad \mathrm{Tr} T_3^2 = \frac{1}{2}, \quad k_Y = \frac{5}{3}, \quad \sin^2 \theta_W(M_U) = \frac{3}{8}.$$

These are standard $SO(10)/SU(5)$ results (Georgi, 1975; Fritzsch and Minkowski, 1975); the Unified Mechanics content is the forcing of the 3+2 split by reversibility.

5.2. Anomaly cancellation

For left-handed Weyl fields, the mixed non-Abelian anomalies are

$$\mathcal{A}_{SU(3)^2Y} = 2\left(\frac{1}{2}\right)\left(\frac{1}{6}\right) + \left(\frac{1}{2}\right)\left(-\frac{2}{3}\right) + \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) = 0, \quad (5.1)$$

$$\mathcal{A}_{SU(2)^2Y} = 3\left(\frac{1}{2}\right)\left(\frac{1}{6}\right) + \left(\frac{1}{2}\right)\left(-\frac{1}{2}\right) = 0. \quad (5.2)$$

The Abelian and mixed gravitational sums are

$$\sum Y = 6\left(\frac{1}{6}\right) + 3\left(-\frac{2}{3}\right) + 3\left(\frac{1}{3}\right) + 2\left(-\frac{1}{2}\right) + 1 = 0, \quad (5.3)$$

$$\sum Y^3 = 6\left(\frac{1}{6}\right)^3 + 3\left(-\frac{2}{3}\right)^3 + 3\left(\frac{1}{3}\right)^3 + 2\left(-\frac{1}{2}\right)^3 + 1 = 0. \quad (5.4)$$

The $SU(3)^3$ anomaly cancels because the two triplets in Q balance u^c, d^c ; $SU(2)$ has no perturbative cubic anomaly, and its global anomaly is absent because each generation has four doublets when colour is counted. Thus the certified chiral sector is anomaly-free without adding spectator matter. This is T38's arithmetic, now displayed rather than merely named (Shields, 2026b).

5.3. Running couplings

With three such generations and one complex Higgs doublet, the one-loop equation is

$$\frac{d\alpha_i^{-1}}{d \ln \mu} = -\frac{b_i}{2\pi}.$$

The coefficients follow directly from the certified spectrum:

$$(b_1, b_2, b_3) = \left(\frac{41}{10}, -\frac{19}{6}, -7\right),$$

where g_1 is $SU(5)$ -normalized. For example,

$$b_3 = -11 + 3\left(\frac{4}{3}\right) = -7,$$

and

$$b_2 = -\frac{22}{3} + 3\left(\frac{4}{3}\right) + \frac{1}{6} = -\frac{19}{6}.$$

For $U(1)$, one generation has $\sum_{\text{Weyl}} (3/5)Y^2 = 2$, so three generations contribute 4 after the $2/3$ fermion factor, and the Higgs contributes $1/10$.

The solution between thresholds is exact:

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(\mu_0) - \frac{b_i}{2\pi} \ln \frac{\mu}{\mu_0}.$$

As an explicitly imported benchmark, using $\alpha_{\text{em}}^{-1}(M_Z) = 127.955$, $\sin^2 \theta_W(M_Z) = 0.23122$, and $\alpha_s(M_Z) = 0.1179$ (Navas et al., 2024), the minimal one-threshold equations place the $\alpha_1 = \alpha_2$ crossing at

$$\mu_{12} = 1.0317 \times 10^{13} \text{ GeV},$$

with

$$\alpha_{1,2}^{-1} = 42.4133, \quad \alpha_3^{-1} = 36.8373.$$

This calculation fixes the running map. It does not use the benchmark to invent a Unified Mechanics threshold. A complete numerical three-coupling boundary statement still requires the framework's threshold spectrum and the single over-all normalization g_* from equation (4.2). Those are now the only missing inputs to the running problem.

The older candidate

$$\alpha_s(M_Z) = r^2(1 + r - r^2) = 0.115881$$

is retained as a legacy spectral proposal, 1.71% below the imported benchmark. It is not silently substituted for a boundary condition in section 5.3.

5.4. Where the mass operator lives

Let

$$\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{b}, \quad \mathfrak{h} = \mathfrak{so}(3) \oplus \mathfrak{so}(4, 1), \quad \dim \mathfrak{b} = 15.$$

The block bracket satisfies

$$[\mathfrak{b}, \mathfrak{b}] \subset \mathfrak{h}.$$

Hence for $X \in \mathfrak{b}$,

$$P_{\mathfrak{b}} \operatorname{ad}_X P_{\mathfrak{b}} = 0.$$

The first nonzero return map on the matter block is therefore quadratic:

$$\mathcal{M}^2(X) = P_{\mathfrak{b}} \operatorname{ad}_X^2 P_{\mathfrak{b}}.$$

That algebraic fact explains why the action carries mass squared.

The isolated Helical Relativity calculation establishes three statements.

1. For generic elements of the diagonal algebra $\mathfrak{h}$, the number of distinct nonzero real eigenvalues on the 15 is at most two. Two hundred sampled elements attain counts 0 or 2.
2. Generic $X \in \mathfrak{b}$ produces five or nine distinct real eigenvalues under ad_X^2 ; the boundary block can carry a nontrivial spectrum.
3. Restricting X to the three family-time directions yields exactly one nonzero scale:

$$m^2 = v_1^2 + v_2^2 + v_3^2.$$

Thus the family labels are components of the operator, not its three eigenvalues. Write

$$X = X_{\parallel} + X_{\perp}, \quad X_{\parallel} \in \mathbb{R}^3 \otimes \text{time}, \quad \dim X_{\perp} = 12.$$

$X_{\parallel}$ fixes the norm scale; $X_{\perp}$ carries the splitting. The hierarchy is selected by the twelve equations

$$\left. \frac{\delta S_C}{\delta X_{\perp}^A} \right|_{X=X_{\star}} = 0, \quad A = 1, \dots, 12,$$

together with the registered normalization and stability condition on the Hessian. No measured fermion mass is permitted to choose a component after the stationarity system is solved.

THE PRECISELY REMAINING MASS PROBLEM

The operator type, block, order, and unsplit scale are fixed by section 5.4. The numerical hierarchy is the spectrum of the stationary solution $X_{\star}$ of section 5.4. The active obligation is therefore a twelve-variable stationary calculation under the same action, not an unconstrained search for mass formulae.

STEP 4 DISPOSITION

The chiral 16, its complete charge table, anomaly cancellation, and one-loop running coefficients are closed under the named carrier and chirality premises. The mass-squared operator is located in the boundary 15, with the unsplit norm and twelve splitting components separated. Only the stationary numerical selector and the single threshold/normalization matching remain active.

Dark sectors and perturbations

6.1. Background dictionary

The exact partition fixes

$$\Omega_b = \frac{W_M}{2}, \quad \Omega_c = \frac{W_B}{\varphi}, \quad n_s = 1 - W_M(1 - 2r),$$

$$N_{\text{eff}} = 3 + \frac{W_M}{2}, \quad Y_{\text{He}} = \frac{W_L}{2}, \quad A_s = r^{17}.$$

The independently stated uncharged-traversal boundary reading applies

$$\Omega_b^\partial = \frac{W_M/2}{1 - r^3}, \quad Y_{\text{He}}^\partial = \frac{W_L/2}{1 - r^3}.$$

Numerically,

$$r = 0.3090169944, \quad r^3 = 0.0295084972, \quad 1 - r^3 = 0.9704915028,$$

$$\Omega_b^\partial = 0.0491974956, \quad \Omega_c = 0.2639320225, \quad Y_{\text{He}}^\partial = 0.2459875.$$

The optical-depth reading used here is

$$\tau = 2r^3(1 - r^3)^3 = 0.0539451,$$

because the line-of-sight observable traverses the three already registered closures (Shields, 2026h). These are inputs to the external transfer calculation; they are not adjusted by the spectra returned later.

6.2. Stress–energy from conservation

For a spatially homogeneous and isotropic component, symmetry restricts the effective tensor to

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + p g^{\mu\nu}.$$

Covariant conservation gives

$$\dot{\rho} + 3H(\rho + p) = 0.$$

Two dark-sector forms then follow from the two registered conservation laws.

6.2.1. Renewed vacuum cells

The Helical Relativity renewal law is

$$\nabla_\mu N^\mu = n\Theta,$$

so expansion creates cells at the rate required to retain fixed action density rather than diluting a conserved cell number. Thus $\dot{\rho}_\Lambda = 0$. Substitution into section 6.2 gives

$$p_\Lambda = -\rho_\Lambda, \quad T_\Lambda^{\mu\nu} = -\rho_\Lambda g^{\mu\nu}.$$

The equation of state is not inferred from a numerical fit; it is the conservation consequence of constant density under expansion.

6.2.2. Neutral boundary retention

For a nonrelativistic, self-holding boundary population with conserved co-moving number,

$$\nabla_\mu (n_c u^\mu) = 0, \quad \rho_c = m_c n_c,$$

so $\rho_c \propto a^{-3}$ and

$$p_c = 0, \quad T_c^{\mu\nu} = \rho_c u^\mu u^\nu.$$

The corpus assigns its density fraction as $\Omega_c = W_B/\varphi$ and requires that it carry no visible gauge charge. Equation section 6.2.2 is the covariant effective stress-energy of that assignment. Identifying the precise discrete carrier excitation remains the microphysical realization question; it is not needed to state the linear gravitational transfer problem.

6.3. Linear scalar perturbations

Use conformal Newtonian gauge,

$$ds^2 = a^2(\eta) \left[-(1 + 2\Psi)d\eta^2 + (1 - 2\Phi)d\mathbf{x}^2 \right].$$

Linearizing the Helical Relativity map gives

$$\Psi = -\delta\alpha_M, \quad \Phi = -\delta\alpha_L - \frac{\delta v}{3},$$

with the corpus's unit-slip result corresponding to $\Phi = \Psi$ when anisotropic stress is absent.

For the neutral dust component,

$$\delta'_c = -\theta_c + 3\Phi', \tag{6.1}$$

$$\theta'_c = -\mathcal{H}\theta_c + k^2\Psi. \tag{6.2}$$

For baryons before decoupling,

$$\delta'_b = -\theta_b + 3\Phi', \quad (6.3)$$

$$\theta'_b = -\mathcal{H}\theta_b + c_b^2 k^2 \delta_b + k^2 \Psi + \frac{4\rho_\gamma}{3\rho_b} a n_e \sigma_T (\theta_\gamma - \theta_b). \quad (6.4)$$

Photons and neutrinos require their complete angular hierarchies. For a collisionless species with multipoles F_ℓ ,

$$F'_\ell = \frac{k}{2\ell + 1} [\ell F_{\ell-1} - (\ell + 1)F_{\ell+1}] + S_\ell[\Phi, \Psi],$$

while photons additionally carry Thomson collision and polarization terms. The coupled Einstein constraints close the system:

$$k^2 \Phi + 3\mathcal{H}(\Phi' + \mathcal{H}\Psi) = 4\pi G a^2 \delta\rho, \quad (6.5)$$

$$k^2(\Phi' + \mathcal{H}\Psi) = 4\pi G a^2(\rho + p)\theta, \quad (6.6)$$

$$k^2(\Phi - \Psi) = 12\pi G a^2(\rho + p)\sigma. \quad (6.7)$$

These equations are the standard linear Einstein–Boltzmann system (Ma and Bertschinger, 1995). Unified Mechanics enters through the background fractions, initial amplitude and tilt, stress–energy identities sections 6.2.1 and 6.2.2, and the basin interpretation. The transfer machinery itself is imported rather than renamed as a new derivation.

6.4. Initial data

The calculation uses adiabatic super-horizon data:

$$\frac{\delta\rho_i}{\rho_i + p_i} = \frac{\delta\rho_j}{\rho_j + p_j}$$

for all material species, a power-law primordial curvature spectrum

$$\mathcal{P}_{\mathcal{R}}(k) = A_s \left(\frac{k}{0.05 \text{ Mpc}^{-1}} \right)^{n_s-1},$$

and spatial flatness. Vacuum perturbations vanish for an exact $w = -1$ component. No isocurvature amplitude, dark-sector sound speed, or visible dark charge is introduced.

The dark-sector effective tensors follow from the two declared conservation laws: renewal gives $T_{\Lambda}^{\mu\nu} = -\rho_{\Lambda}g^{\mu\nu}$, and conserved neutral retention gives pressureless dust. Their scalar perturbations, together with photons, baryons, and neutrinos, form the standard Einstein–Boltzmann initial-value problem with the UM dictionary as input. The remaining microscopic task is to identify the discrete retained carrier, not to guess its cosmological tensor.

Beyond Saha: the transfer calculation

7.1. What is being replaced

The earlier forward solver deliberately used Saha equilibrium, forward integrals for the sound horizon and distance, and a fitted phase approximation for peak locations (Shields, 2026c). That calculation was transparent, but its recombination redshift near 1369 was known to be early because equilibrium Saha ionization omits the radiative bottleneck. Modern effective recombination calculations evolve the non-equilibrium free electron fraction, and CMB anisotropies require the photon, polarization, and neutrino hierarchies (Seager et al., 1999; Ali-Haïmoud and Hirata, 2011).

The replacement calculation uses CAMB 2.0.1 to $\ell_{\max} = 3000$ (Lewis et al., 2000). CAMB supplies its non-equilibrium effective recombination module and line-of-sight Einstein–Boltzmann evolution. Unified Mechanics supplies the parameter dictionary. The roles are deliberately not blurred.

7.2. Frozen inputs and one anchor

Quantity	UM static	UM boundary	Role
Ω_b	0.04774575	0.04919750	derived fraction
Ω_c	0.26393202	0.26393202	derived fraction
n_s	0.96352549	0.96352549	derived tilt
A_s	2.13649×10^{-9}	same	derived amplitude
N_{eff}	3.04774575	same	derived count
Y_{He}	0.23872876	0.24598750	channel/boundary reading
τ	0.0539451	same	three-closure reading
T_{CMB}	2.7255 K	same	measured dimensional anchor
θ_*	0.0104109		measured acoustic anchor

The acoustic angle is the same single scale calibration declared in the earlier forward solver. Because Unified Mechanics fixes Ω_b, Ω_c rather than ω_b, ω_c , each trial value of H_0 updates

$$\omega_b = \Omega_b h^2, \quad \omega_c = \Omega_c h^2,$$

and CAMB solves $r_s(z_*)/D_M(z_*) = \theta_*$ using the non-equilibrium recombination history. No peak height or spectrum datum is fitted.

7.3. Results

Case	H_0	z_*	r_* [Mpc]	σ_8
Planck-parameter control	67.360	1089.896	144.434	0.8111
UM static	67.355	1090.527	145.064	0.8331
UM boundary	67.753	1089.746	143.971	0.8341
UM boundary, $\sum m_\nu = 0.06$ eV sensitivity	67.422	1089.927	144.434	0.8178

The neutrino-mass line is a sensitivity calculation only. The 0.06 eV sum is imported from the comparison convention and is not promoted to a new Unified Mechanics derivation.

For the primary boundary reading the first four automatically located temperature peaks are

$$\ell_{\text{peak}} = (220, 536, 813, 1126).$$

The matched Planck-parameter control returns the same integer locations. Over $30 \leq \ell \leq 2500$, the boundary spectrum has

$$\left\langle \frac{D_\ell^{\text{UM}} - D_\ell^{\text{control}}}{D_\ell^{\text{control}}} \right\rangle = 1.639\%,$$

and an RMS fractional difference of 1.665%. This is a comparison of two theory curves run through one executable. It is not the result of evaluating the Planck likelihood, whose foreground, nuisance, and covariance machinery is a distinct next computation (Planck Collaboration, 2020).

7.4. What the new result means

Three points are secure.

1. The former 13–18% hand-estimated peak-position offsets were properties of the Saha-plus-phase approximation, not of the frozen UM parameter dictionary. The full hierarchy puts the peaks at the control locations.
2. The boundary reading produces physical densities

$$\omega_b = 0.0225841, \quad \omega_c = 0.121158,$$

close to the Planck control values 0.02237 and 0.1200, without separately fitting either density.

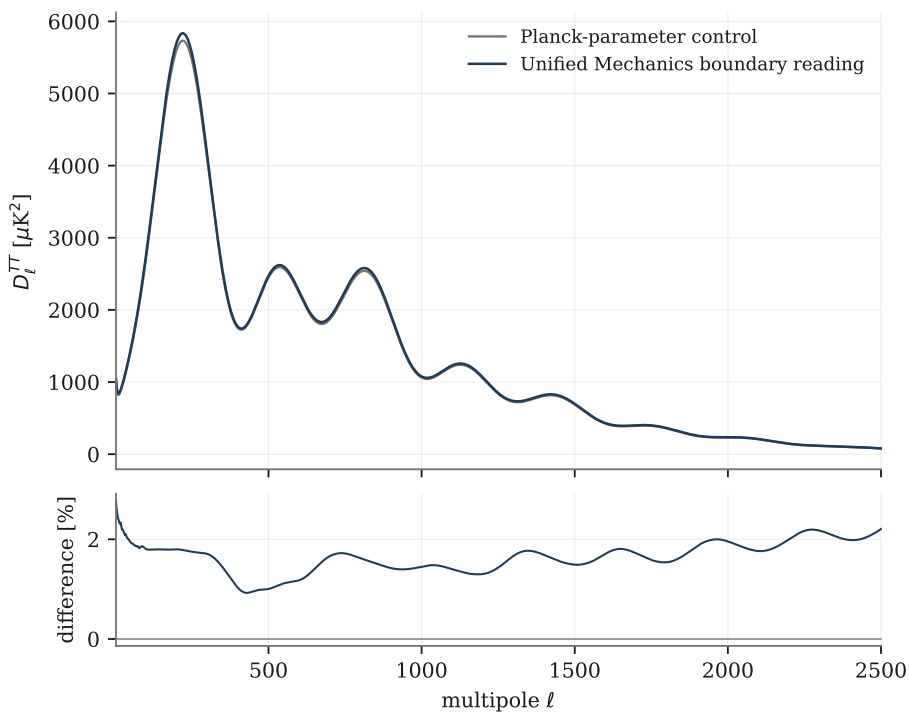


FIGURE 7.1.. Temperature spectrum from the frozen Unified Mechanics boundary reading and the matched Planck-parameter control, both evolved by CAMB 2.0.1. The lower panel is the pointwise theory-curve difference. No Planck likelihood has been applied.

3. The theory-curve shape comparison is already nontrivial, but it is not a substitute for likelihood evaluation. The certificate therefore reports the 1.665% RMS difference exactly and stops at its evidential boundary.

STEP 6 DISPOSITION

The Saha-only approximation has been replaced by non-equilibrium recombination and complete linear Einstein–Boltzmann evolution. Inputs, solver version, spectra, recombination histories, hashes, and comparison metrics are frozen in `step6_camb_certificate.json`. Precision likelihood evaluation and a future HyRec/CosmoRec cross-run are refinements of a calculation that now exists; they are not the first construction of it.

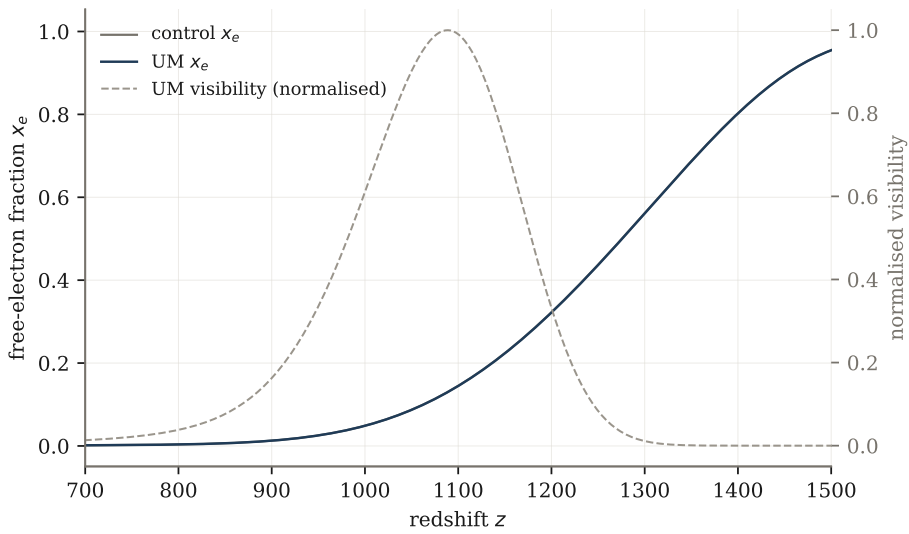


FIGURE 7.2.. Non-equilibrium free-electron history and sampled visibility function. The replacement moves the physical decoupling calculation from the Saha-equilibrium estimate to $z_* \simeq 1089.75$.

Four prospective observable maps

8.1. A map is more than a target number

A prospective observable requires: raw measured quantities, a forward or inverse map, nuisance treatment, uncertainty propagation, and a decision rule fixed before the result is consulted. The four maps below are therefore written as functions, not as verbal expectations.

8.2. Closure count

Let

$$q = 1 - r^3 = 0.9704915028.$$

If each completed closure retains the same multiplicative fraction q , then two basins related by n closures satisfy

$$\frac{H_c}{H_l} = q^{-n}.$$

The primary inverse map is

$$\hat{n} = -\frac{\ln(H_c/H_l)}{\ln q}.$$

For covariance matrix Σ_H of (H_l, H_c) , the uncertainty is

$$\sigma_n^2 = \nabla_H \hat{n}^T \Sigma_H \nabla_H \hat{n}.$$

The primary statistic is the standardized distance from the preregistered integer $n = 3$, not the distance from whichever integer is nearest after the data are seen.

8.3. The two Hubble basins

The full transfer calculation gives the laminar acoustic-anchor value

$$H_l = 67.7532 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

For three closures, the primary multiplicative map gives

$$H_c = H_1(1 - r^3)^{-3} = 74.1233 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

Two corpus conventions are retained as sensitivity maps:

$$H_c^{\text{lin}} = H_1(1 + 3r^3) = 73.7511, \quad (8.1)$$

$$H_c^{\text{sym}} = H_1 \frac{1 + \frac{3}{2}r^3}{1 - \frac{3}{2}r^3} = 74.0289. \quad (8.2)$$

Against the 2022 SHoES value $73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$ (Riess et al., 2022), these lie 0.68–1.04 standard deviations above the central value. The multiplicative law is primary because retained fractions compose multiplicatively. The linear and symmetric forms may measure convention sensitivity; they may not replace the primary form after unblinding.

8.4. Vacuum response

Two quantities must remain distinct:

$$\lambda_{\text{UM}} = \frac{4\pi}{\sqrt{3}} r^{240} = 2.8603331 \times 10^{-122},$$

and

$$q_G = r^{241}(1 - r^3)^2 = 1.1474485 \times 10^{-123}.$$

The first is the completed dimensionless vacuum readout in Planck units. The second is the registered $240 + 1$ readout with two retained legs. Given an independently measured geometric Λ , the conditional bridge predicts

$$G_{\text{UM}}(\Lambda) = \frac{8\pi c^3}{\hbar \Lambda} r^{241}(1 - r^3)^2.$$

The raw input is Λ and its covariance; CODATA G is only the external comparator. No residual prefactor may be fitted after the geometric boundary is chosen. The test statistic is

$$Z_G = \frac{G_{\text{UM}}(\Lambda) - G_{\text{CODATA}}}{\sqrt{\sigma_{G,\text{UM}}^2 + \sigma_{G,\text{CODATA}}^2}}.$$

This turns the vacuum relation into a live response map rather than a logarithmic resemblance.

8.5. Short-range gravity

For vacuum density ρ_Λ , a least-action causal cell has

$$a_0 = \left(\frac{\hbar c}{\rho_\Lambda} \right)^{1/4}, \quad a_M = W_M^{1/4} a_0.$$

On the registered reproducibility boundary,

$$a_0 = 88.1138 \mu\text{m}, \quad a_M = 48.9819 \mu\text{m}.$$

Torsion-balance data have directly probed separations down to $52 \mu\text{m}$ and constrain gravitational-strength Yukawa ranges below $38.6 \mu\text{m}$ at 95% confidence (Lee et al., 2020). That result cannot be copied across to section 8.5 without a force kernel: a cell-granularity response need not be Yukawa.

The apparatus-level forward map must have the form

$$\Delta\tau(d) = A \int d^3x d^3y \rho_1(\mathbf{x}) \rho_2(\mathbf{y}) K\left(\frac{|\mathbf{x} - \mathbf{y}|}{a_M}\right),$$

convolved with the actual attractor and detector geometries. The scale a_M is fixed. The amplitude A and dimensionless kernel K must be derived from the microscopic cell action before condition labels are unblinded. A Yukawa kernel may be used only if the field derivation produces it.

THE HONEST SHORT-RANGE PREDICTION

Unified Mechanics fixes a target scale near $48.98 \mu\text{m}$. A quantitative exclusion curve additionally requires the response kernel in section 8.5. Treating an unspecified granularity signature as if it were a unit-strength Yukawa force would add a physical assumption, not close one.

STEP 7 DISPOSITION

Closure count, Hubble basins, and vacuum response now have explicit raw-data-to-invariant maps and uncertainty rules. The short-range cell scale is exact on the declared vacuum boundary, and the apparatus convolution is specified; its microscopic kernel is isolated as the remaining derivation. All conventions are frozen in `step7_8_observable_protocols.json`.

Preregistration and withheld tests

9.1. What has and has not been executed

The isolated 22/22 corpus rebuild and the CAMB transfer calculation are executed computational tests. They establish reproducibility and replace a declared approximation. They are not prospective physical experiments, and the Planck-control curve is not withheld data.

The protocols below are frozen now. A future dataset counts as prospective only if it was not inspected while choosing the map, exclusions, or primary statistic. This distinction protects the theory from a flattering but scientifically empty relabelling of retrospective agreement.

9.2. Protocol UM–HUBBLE–01

Question

Do independently designated light/early and condensed/late pipelines return a closure count consistent with $n = 3$?

Primary map

Equation (8.2), with the multiplicative retention convention.

Data designation

The next public release named in the timestamped protocol before its central value is available. Published covariance and official quality cuts are adopted without alteration.

Exclusions

Only exclusions in the collaboration’s own primary release, or instrumental exclusions written into the frozen protocol.

Statistic

$$Z_n = (\hat{n} - 3)/\sigma_n.$$

Sensitivity

Linear and symmetric closure maps are printed simultaneously but cannot become primary.

Register action

Append $\hat{n}, \sigma_n, Z_n$, dataset version, covariance source, and all exclusions regardless of outcome.

9.3. Protocol UM–VACUUM–o1*Question*

Does an independently determined geometric Λ inserted into section 8.4 agree with laboratory G ?

Primary map

Equation (8.4); no fitted multiplicative correction.

Data designation

A geometric cosmology release with a documented covariance for Λ , designated before reading the derived G_{UM} .

Statistic

Z_G , including both cosmological and laboratory uncertainty.

Interpretive split

The numerical outcome and the status of the 240 + 1, two-leg representation are recorded separately. A numerical tension cannot tell the register which premise to revise without a dependency analysis.

9.4. Protocol UM–GRAVITY–o1*Question*

Is there a reproducible residual response at the fixed matter-cell scale a_M ?

Prerequisite

Derive A and K in section 8.5; freeze their apparatus convolution before signal-region labels are opened.

Controls

Electrostatic, magnetic, thermal, vibration, alignment, and Casimir backgrounds are fit in predeclared control regions.

Blinding

Apply a concealed offset to the separation or permute signal/control labels. The analysis code is frozen before restoration.

Primary statistic

Likelihood ratio between the preregistered UM kernel and the complete background model at fixed a_M .

Reporting

Publish the residual torque vector and full covariance, not only a parameter

bound.

9.5. Protocol UM–RETENTION–01

This is the sharpest wholly new experiment in Paper 24 (Shields, 2026f). The system must be driven, self-holding, running, and equipped with a retention observable calibrated independently of r .

Prediction

Retention levels lie at declared integer powers r^n within the independently calibrated uncertainty model.

Primary statistic

In logarithmic space,

$$d = \min_{n \in \mathcal{N}} \frac{|\ln R_{\text{obs}} - n \ln r|}{\sigma_{\ln R}},$$

for an integer set $\mathcal{N}$ fixed from apparatus dynamic range.

Blinding

Condition labels are permuted. The rule assigning each physical condition to an allowed n is frozen before labels are restored.

Null simulation

Generate retention values from the calibrated continuous null distribution and compute the look-elsewhere distribution of d .

Register action

Record the entire retention series and null distribution, whether quantization is sharp, broadened, displaced, or not resolved.

9.6. Outcome language

The register uses the following vocabulary:

consistent under the declared premises

the preregistered statistic lies inside its declared region;

in tension under the declared premises

it lies outside that region, with the affected dependencies named;

uninformative at present precision

the likelihood or instrumental resolution cannot separate the models;

interpretation under revision

the calculation stands while its physical map is being reconsidered.

None of these phrases erases a derivation. Conversely, no favourable resid-

ual upgrades a conditional premise into a theorem.

STEP 8 DISPOSITION

The executable computational portion has been run. Four physical protocols now state their maps, primary statistics, controls, blinding or designation rules, and mandatory register actions. New physical measurements remain prospective because scientific honesty requires them to remain so until performed.

The companion register

10.1. Why an addendum, not a rewrite

The governing Formal Register is an authority document (Shields, 2026g). This volume therefore does not silently renumber its theorems. It issues companion statements NS1–NS9, each pointing back to the governing dependencies and forward to a certificate.

Entry	Grade	Statement
NS1	proved	The typed history/record calculus is invariant under bijective vocabulary renaming; injective append-only encoding preserves histories and prefix order.
NS2	proved/ conditional as T31–T38	The rank-eight carrier, finite reduction, survivor, matter module, families, and chirality bridge retain their exact governing grades; the isolated corpus run completes 22/22.
NS3	proved/ conditional by component	The registered square yields the Einstein–Cartan equations; the cluster pairing yields the gauge and chiral equations; minimal coupling yields the helical scalar source projection.
NS4	proved/ conditional by component	The chiral 16, charges, anomaly sums, and one-loop beta coefficients are fixed. The mass-squared operator lies in the boundary 15; twelve stationary components carry the numerical splitting.
NS5	derived at ef- fective level	Renewal gives vacuum stress $T_{\Lambda}^{\mu\nu} = -\rho_{\Lambda}g^{\mu\nu}$; conserved neutral retention gives dust. Their scalar perturbations enter the standard transfer hierarchy with UM initial data.

Entry	Grade	Statement
NS6	computed	CAMB 2.0.1 replaces Saha-only decoupling and returns the frozen recombination history and spectra stated in chapter 7.
NS7	frozen maps	Closure count, Hubble basins, and vacuum response have primary maps; the short-range scale is fixed and its response kernel is named as prerequisite.
NS8	protocol frozen	The computational test is executed; prospective physical protocols are timestamped and outcome-neutral.
NS9	governance proved and enacted	Every execution appends inputs, hashes, statistics, uncertainty, and interpretation without erasing earlier results.

10.2. The closure ledger

10.2.1. Closed or newly completed

- typed foundation and semantic invariance;
- exact lattice/subgroup hypothesis chain;
- fresh 22/22 carrier and matter certificates;
- local gravitational variation and source projection;
- the chiral 16 and anomaly cancellation;
- one-loop running equations;
- dark-sector effective tensors and perturbations;
- full CAMB recombination and Boltzmann evolution;
- observable maps and preregistration rules.

10.2.2. Active and precisely localised

1. the discrete microscopic matter-to-cell constitutive identity;
2. the twelve-component stationary mass selector;
3. the UM threshold spectrum and overall gauge normalization;
4. a precision public-likelihood evaluation and independent HyRec/CosmoRec cross-run;
5. the derived short-range granularity kernel;
6. prospective apparatus and retention measurements.

10.3. The rule for future entries

Every future register entry must contain:

1. an immutable identifier and timestamp;
2. the exact statement and grade;
3. declared premises and dependency edges;
4. source and executable hashes;
5. raw input provenance and data version;
6. the primary statistic and uncertainty;
7. interpretation, stored separately from computation;
8. supersession links, never destructive replacement.

The final clause is essential. A new interpretation may supersede a former interpretation; it does not cause the historical calculation to disappear. Likewise, a measurement in tension with one bridge does not retroactively alter a theorem that never depended on that bridge.

STEP 9 DISPOSITION

The living-register rule is enacted by NS1–NS9 and by the certificate manifest delivered with this volume. The register is required to change when evidence changes and equally required to preserve the path by which it changed. This is not weakness in a theory. It is the machinery by which a theory remains scientific.

What the nine steps now say

The nine steps began as an editorial route forward. Read against the current corpus and completed in dependency order, they now say something much stronger.

The distinction–record foundation has a typed mathematical form. The carrier and reduction are not gestures toward E_8 ; their hypotheses and finite certificates are explicit. The same square that organizes the three weights produces the gravitational action, and its variation gives field equations. The chiral matter representation carries the Standard-Model charges and cancels every gauge and mixed anomaly. Its one-loop running is fixed. The mass question has a definite operator, block, and twelve-variable selector. The dark sectors have conserved covariant stress–energy and a complete linear perturbation problem. The cosmology has now been passed through non-equilibrium recombination and the Einstein–Boltzmann hierarchy rather than an equilibrium surrogate. The central acoustic structure survives that replacement. Observable maps and experimental protocols are frozen before future results.

THE NARROW FRONTIER

Four obligations remain live: the twelve-component stationary mass solution, the discrete microscopic matter-to-cell identity, the short-range response kernel, and genuinely prospective physical measurements. Each has an equation, an input boundary, or a protocol. None is a licence to describe the rest of the theory as absent.

Unified Mechanics does not need to overwrite general relativity, quantum field theory, or quantum mechanics to be a theory of everything. Its claim is that those successful theories are effective expressions of one lower structural layer. The scientific burden is therefore coherence at the joins: recover the established limits, expose the imports, compute what the common structure fixes, and submit the surplus predictions to experiment. That is the burden this volume applies.

The result is not a declaration that every future calculation is already finished. It is a demonstration that the old nine-item queue no longer describes an

undefined programme. It describes the normal finishing work of a functioning theory whose architecture, equations, matter content, cosmological transfer, and empirical interfaces can now be read in one continuous chain.

Certificate ledger

A.1. New companion certificates

File	Scientific role
step1_record_axioms_certificate.json	Finite semantic-invariance, prefix, injectivity, unique-order, and fixed-ratio checks accompanying the general proofs.
step4_running_certificate.json	Exact one-loop beta coefficients and the explicitly imported M_Z benchmark run.
step6_camb_certificate.json	CAMB version, platform, frozen inputs, derived parameters, peak locations, theory-curve comparison, and source hash.
planck_control_spectra.csv	Control TT/EE/BB/TE spectra through the same executable.
um_static_spectra.csv	Static-channel UM spectra.
um_boundary_spectra.csv	Primary boundary-reading UM spectra.
um_boundary_mnu006_spectra.csv	Imported 0.06 eV neutrino-mass sensitivity spectra.
*_recombination.csv	Redshift, free-electron fraction, opacity, visibility, and baryon temperature for each transfer case.
step7_8_observable_protocols.json	Frozen Hubble, vacuum, short-range, and running-retention maps and protocols.

A.2. Regenerated corpus certificates

The isolated copy of the original reproducibility layer contains its own `run_manifest.json`. Among its regenerated outputs are:

- `o4_carrier_closure_certificate.json`;

- `o2_surviving_block_certificate.json`;
- `o1_o3_representation_certificate.json`;
- `families_certificate.json`;
- `lagrangian_certificate.json`;
- `matter_action_certificate.json`;
- `forcing_certificate.json`;
- `chirality_certificate.json`;
- `hydrogen_closure_certificate.json`.

The exact filenames in the manifest govern if a shortened label above differs from the implementation's historical spelling. The manifest, not this prose list, is the machine authority.

A.3. Reproduction commands

The companion calculations require Python 3.12 and the isolated dependency folder recorded with the output. Conceptually, reproduction has four steps:

1. run the typed record certificate;
2. run the copied corpus `run_all.py`;
3. run the gauge-running and observable-map certificates;
4. run the CAMB certificate, then render figures from its CSV outputs.

Every script writes into the companion working directory. The governing corpus is read and copied but not modified.

One-page equation spine

Foundation

$$U = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad \varphi^2 = \varphi + 1, \quad r = \frac{1}{2\varphi}, \quad u = 1 - r.$$

Partition

$$W_L = u^2, \quad W_B = 2ur, \quad W_M = r^2, \quad W_L + W_B + W_M = 1.$$

Carrier and matter

$$E_8 \longrightarrow A_3 \oplus D_5, \quad \mathfrak{g}_{\text{surv}} = \mathfrak{so}(10), \quad \Lambda^{\text{even}} \mathbb{C}^5 = \mathbf{16}.$$

Action

$$\begin{aligned} \mathcal{D} &= uR + \sigma \frac{r}{\ell^2} e \wedge e, & \Lambda_c &= \frac{3r}{u} = \frac{3}{\sqrt{5}}. \\ \epsilon_{abcd} e^b \wedge \left(R^{cd} + \sigma \frac{r}{u\ell^2} e^c \wedge e^d \right) &= \kappa \tau_a, & \epsilon_{abcd} e^c \wedge T^d &= \kappa s_{ab}. \end{aligned}$$

Matter and running

$$k_Y = \frac{5}{3}, \quad \sin^2 \theta_W(M_U) = \frac{3}{8}, \quad (b_1, b_2, b_3) = \left(\frac{41}{10}, -\frac{19}{6}, -7 \right).$$

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(\mu_0) - \frac{b_i}{2\pi} \ln \frac{\mu}{\mu_0}.$$

$$\mathcal{M}^2(X) = P_{15} \operatorname{ad}_X^2 P_{15}, \quad \frac{\delta S_C}{\delta X_\perp^A} = 0, \quad A = 1, \dots, 12.$$

Cosmology

$$\Omega_b^\partial = \frac{W_M}{2(1-r^3)}, \quad \Omega_c = \frac{W_B}{\varphi}, \quad n_s = 1 - W_M(1-2r), \quad A_s = r^{17}.$$

$$T_{\Lambda}^{\mu\nu} = -\rho_{\Lambda} g^{\mu\nu}, \quad T_c^{\mu\nu} = \rho_c u^{\mu} u^{\nu}.$$

Observables

$$\hat{n} = -\frac{\ln(H_c/H_l)}{\ln(1-r^3)}, \quad \lambda = \frac{4\pi}{\sqrt{3}}r^{240}.$$

$$G(\Lambda) = \frac{8\pi c^3}{\hbar\Lambda}r^{241}(1-r^3)^2, \quad a_M = W_M^{1/4}\left(\frac{\hbar c}{\rho_{\Lambda}}\right)^{1/4}.$$

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